International Conference on Semigroups and Applications

Abstracts

14th–18th September 2026, University of Manchester

Celebrating the 65th birthday of Professor Victoria Gould

Contents

Plenary talks

Tom Aird

Tuesday 15th, 09:00

University of Manchester

Maximal subgroups of tropical matrix monoids

The set of n-by-n matrices over the tropical semiring, under matrix multiplication, forms a monoid with many interesting algebraic and combinatorial properties. In this talk, we investigate the maximal subgroups of these tropical matrix monoids.

Unlike the classical case over a field, tropical matrices have much weaker notions of rank, giving rise to an intricate lattice of $\mathcal{J}$-classes determined by the row and column spaces of the underlying matrices. I will begin with an introduction to tropical matrices, tropical modules, and the Green's relations of the monoid, before describing how the maximal subgroups can be identified: they arise as automorphism groups of associated combinatorial structures. I will then present a full classification of the maximal subgroups of the monoid of n-by-n tropical matrices, comparing our results to related work in the literature. Finally, I will discuss how these results extend to Schützenberger groups and to the category of non-square matrices.

João Araújo

Thursday 17th, 09:00

Universidade Nova de Lisboa

Complete mappings for semigroups

A complete mapping of a semigroup S is a bijection $\alpha\colon S \to S$ such that the map $\theta\colon S \to S$ defined by $x\theta = x \cdot x\alpha$ is also a bijection. Complete mappings connect group theory, Latin squares, and cryptography, and their existence for finite groups was characterized by the resolution of the Hall–Paige conjecture. Here I will develop the corresponding theory for finite semigroups.

We prove that every finite semigroup admitting a complete mapping is regular and that the problem reduces to principal factors. We classify the existence of a complete mapping in Rees matrix semigroups without zero, give a Hall-type criterion for Rees 0-matrix semigroups over groups with complete mappings, and prove sufficient conditions for Rees 0-matrix semigroups whose maximal subgroups do not have complete mappings. As the main application of the Rees 0-matrix analysis, we show that T_n has a complete mapping if and only if the same holds for S_n. The proofs combine Green–Rees structure theory with the Hall–Paige theorem and Burnside transfer, the Hall–Gale–Edmonds framework for matchings, flows, matroids and polyhedra, Hoffman–Kruskal total unimodularity and integral polyhedra, Kőnig edge-colouring, Bevis–Hall–Katz incidence theory over finite abelian groups, and Bregman–Egorychev–Falikman permanent estimates.

Robert Gray

Friday 18th, 09:00

University of East Anglia

Title to be confirmed.

Abstract to be confirmed.

Christopher Hollings

Monday 14th, 15:15

University of Oxford

Beyond semigroups: the mathematical life of A. K. Sushkevich

In this talk, I will outline the wider work and biography of a figure whose name, but perhaps not much else about him, is well known to semigroup theorists: A. K. Sushkevich. Beginning with a discussion of his studies in Berlin at the start of the twentieth century, I will then describe the academic career that he built in Soviet Ukraine. As well as summarising his work (on semigroups) as a research mathematician, I will also consider his mathematical teaching, with a focus on how he taught group theory.

Ganna Kudryavtseva

Wednesday 16th, 16:00

University of Ljubljana

Proper Ehresmann semigroups

The talk will present the notion of a proper biEhresmann semigroup which generalizes that of a proper birestriction semigroup. Because of the absence of the ample identities, this notion is unavoidably much more complex than that of a proper birestriction semigroup. Our underlying idea is to extend the presentation of proper birestriction semigroups as partial action products of a semilattice by a monoid.

Starting from a semilattice E, a monoid T and a directed graph $\mathcal{G}$ possessing additional structure that we call compatible restrictions and corestrictions subject to a certain condition, with vertex set E and edges labelled by elements of T, we construct a biEhresmann semigroup $E \rtimes_{\mathcal{G}} T$. This semigroup turns out to be our desired generalization, whence we term so arising biEhresmann semigroups proper.

We show that every biEhresmann semigroup possesses a proper cover and, remarkably, whenever S is a biEhresmann monoid with projections $P(S)$, the covering monoid $P(S) \rtimes_{\mathcal{G}} X^*$ is isomorphic to the monoid $\mathcal{P}(X^*, P(S))$, defined using a different approach, from the work of Branco, Gomes and Gould.

This is joint work with Valdis Laan.

António Malheiro

Tuesday 15th, 16:30

NOVA University of Lisbon

Title to be confirmed.

Abstract to be confirmed.

Craig Miller

Wednesday 16th, 09:00

Durham University

Right noetherian semigroups and finite generation

A semigroup is said to be right noetherian if all its right congruences are finitely generated. In this talk, we examine the fundamental open problem of whether every right noetherian semigroup must be finitely generated. In particular, we discuss recent developments in the case of cancellative semigroups.

Carl-Fredrik Nyberg-Brodda

Thursday 17th, 16:30

Korea Institute for Advanced Study

Profinite rigidity and finite shadows of infinite objects

When is a finitely generated algebraic object determined by its finite quotients? This is the core question underlying profinite rigidity. I will give an introduction to some of the key ideas behind profinite rigidity, and discuss some recent results concerning the non-rigidity of some groups. In particular, I will show that braid groups are not profinitely rigid, giving the first known examples of profinitely non-rigid mapping class groups. I will also give a brief overview of a recent result that free-by-cyclic groups are not relatively profinitely rigid, which answered a question of Bridson & Reid from 2015. Time permitting I will also discuss what happens in the semigroup setting, especially for free semigroups.

Nik Ruškuc

Monday 14th, 13:00

University of St Andrews

Groups arising from idempotent structure

Joint work, and underlying friendship, with Vicky Gould have been one of the aspects of my academic journey that I value most. Among the diverse topics that the two of us have shared, the one in this talk is unique in that we have spent a lot of time talking about it but never actually worked together. In this talk I will try to continue this conversation, have our mutual mathematical friends join in, and extend the invitation to anyone else who is interested. The objects in focus are the so-called free idempotent generated semigroups, which arise from abstracting the essential structure of idempotents (e.g. via biordered sets), together with their counterparts in regular $*$-semigroups, the free projection-generated semigroups. In each case, the properties and structure of these free objects are to a large extent, but not entirely, governed by their maximal subgroups, and the question arises to investigate this relationship. I will try to present the background in non-technical terms, and then concentrate on some recent results, obtained jointly with East, Gray and Mohammed, which put the two types of free objects side by side for comparison and contrast, both in general terms, and also in the special case of partition and other diagram monoids. I will conclude with open problems which to my mind are most important or promising to address, and hope that Vicky and I will get to work on some of them in the days to come.

Dandan Yang Friday 18th, 11:30

Xidian University

Equations, purity and coherency

In this talk, I will discuss my joint work with Vicky Gould concerning the following long-standing and intriguing problem: when does the guaranteed solvability of every finite consistent system of equations in one variable lift to the guaranteed solvability of every finite consistent system of equations in any finite number of variables? This question has a positive answer for some algebraic structures, such as groups and semigroups, but remains far from fully understood for modules over rings or acts over monoids.

Contributed talks

Josiah Aakre

Friday 18th, 11:00

University of Manchester

Katsura-Exel-Pardo groupoids and vanishing of the singular ideal

We give the first examples of non-contracting self-similar groupoids for which simplicity of the Steinberg and C^*-algebras built from the associated ample groupoid is decidable. Katsura-Exel-Pardo (KEP) groupoids find their origin in Katsura's model, built from two integer matrices, which can realize all Kirchberg C^*-algebras. Exel and Pardo later realised these matrices describe a self-similar action of $\mathbb{Z}$ on a directed graph. To avoid the potential perils of topological groupoids and norm completions, we choose to navigate the friendlier landscape of inverse semigroup algebras. The main results are achieved by describing the ideal in question in terms of the self-similar action which, in the setting of KEP-groupoids, may be understood quite well even for non-contracting actions.

Wajih Ashraf

Thursday 17th, 14:00

Aligarh Muslim University, India

Applications of Isbell zigzag theorem

Fennemore, in [7], had described the lattice of all varieties of bands and given its diagram. After that, Petrich [10, Theorem II.5.1] had classified a semigroup identity on bands in at most three variables. Though the varieties of semilattices and left (right) zero semigroups are absolutely closed, the varieties of rectangular bands and right (left) normal bands are not absolutely closed (see Higgins [8, Chapter 4]). Therefore, it is worth finding subvarieties of the variety of all semigroups that are closed in itself or closed and saturated in the containing varieties of semigroups. Encouraged by the fact that Scheiblich [11] had shown that the variety of all normal bands was closed, Alam and Khan in [4–6] had shown that the varieties of all left [right] regular bands, left [right] quasi-normal bands and left [right] semi-normal bands were closed. In [2], Ahanger and Shah had proved a stronger fact that the variety of all left [right] regular bands was closed in the variety of all bands and left [right] semi-normal bands was saturated in bands. In [3], Ahanger, Nabi and Shah had proved the facts that the variety of all rectangular bands was closed in the variety of all bands, the variety of all regular bands was closed and the variety of generalized left [right] regular semigroups was saturated. Finally in [1], Abbas, Ashraf and Khan extended the work in the same direction and proved that the variety of all rectangular bands is closed in the variety of n-nilpotent extension of bands and the variety of normal bands was closed in the variety of left [right] semiregular bands. They also provided a simple and shorter proof of closedness of variety of regular bands. The Isbell zigzag theorem, which is a most useful characterization of semigroup dominions provided by Isbell [9], has played the central role in determination of closed and saturated varieties of semigroups. So, in this way we worked on the applications of Isbell Zigzag theorem. In the same direction, some more closed and saturated varieties are determined. However a complete determination of closed and saturated varieties still remains an open problem. For example, are left (right) semiregular bands closed and regular bands saturated? The affirmative answers of these problems will definitely prove that epis are onto for these classes which, in itself, is an important question.

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[4] Alam, N., Khan, N. M.: Special semigroup amalgams of quasi unitary subsemigroups and of quasi normal bands, Asian European Journal of Mathematics (2013).

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Yakiv Baiduk Thursday 17th, 11:00

Kyiv School of Economics

Transformation semigroup perspective on the magma monoid

We study the semigroup structure on the set of binary operations on a fixed set, which we refer to as the magma monoid. Using a new transformation-semigroup framework, we characterize principal left and right ideals, describe Green's relations, idempotent and regular elements in this structure. In addition, we derive explicit combinatorial enumerations for these classes of elements.

Reinis Cirpons

Wednesday 16th, 11:00

Nantes Université, INRIA

Extensions of the Todd-Coxeter algorithm via automata theory

The well known Todd-Coxeter algorithm is widely used to construct transformation representations of finitely presented monoids. It can be seen, in some sense, as computing the direct limit of a sequence of semiautomata approximating the right Cayley graph of the underlying monoid. This algorithm can also be used for finitely presented groups and, perhaps more surprisingly, there exists an analogue of the Todd-Coxeter algorithm for inverse monoids due to J. B. Stephen (1987). This is despite the fact that the free inverse monoid is not finitely presented as a monoid. Could it be possible to extend the Todd-Coxeter algorithm to other classes of monoids, such as finitely based varieties of monoids?

In this talk I will briefly describe a category theory based approach to modeling the Todd-Coxeter algorithm and I will present an extension of the Todd-Coxeter algorithm to presentations in finitely based varieties of monoids, motivated by a recent NL-space algorithm for deciding if a monoid identity is modelled by a given finitely generated transformation monoid due to L. Fleischer and T. Jack (2020).

Gemma Crowe

Tuesday 15th, 11:00

University of Manchester

Simultaneous conjugacy in Thompson's group V

The simultaneous conjugacy problem asks if given a pair of tuples $(g_1, \ldots, g_n), (h_1, \ldots, h_n) \in G^n$ where G is a group, does there exist a conjugator $x \in G$ such that $(g_i)^x = h_i$ for all $1 \leq i \leq n$. In this talk, we discuss a soon to be released arXiv paper which shows this decision problem is solvable in Thompson's group V. Based on joint work with Luna Elliott.

Levent Michael Dasar

Tuesday 15th, 14:30

University of York

Chain conditions on Green's posets

Let S be a semigroup. The partition of an S-act by its $\mathcal{R}$ or $\mathcal{J}$-relation can be naturally equipped with a partial ordering. We begin by examining what can be said about chain conditions on these posets from an order-theoretic perspective, before giving exact conditions for the preservation of chain conditions under direct products.

James East

Tuesday 15th, 10:00

Western Sydney University

Matrix representations of diagram monoids and categories

The partition monoid P_n is an important algebraic object that arises in many parts of mathematics and science. It contains well known submonoids such as the Brauer monoid B_n and the Temperley–Lieb monoid TL_n. We show that P_n has a faithful 2^n-dimensional representation by zero-one matrices over any additively-idempotent semiring. In the case of the Boolean semiring, this is equivalent to a representation by binary relations. The restrictions to B_n and TL_n contain faithful sub-representations of lower dimension; for TL_n the dimension is the nth Fibonacci number. Interpreting the zero-one matrices instead over a ring of characteristic 0 or an appropriate power of 2, one obtains faithful representations of the (finite or infinite) twisted versions of these monoids. In fact, all of the above can be formulated in the context of diagram/matrix categories, reflecting several other structures such as natural orders and tensor operations. This is all joint work with Marianne Johnson and Mark Kambites (University of Manchester).

Matthias Fresacher

Tuesday 15th, 15:00

Categorical representation of DRC-semigroups

DRC-semigroups model associative systems with domain and range operations, and contain many important classes, such as inverse, restriction, Ehresmann, regular $*$-, and $*$-regular semigroups. In this joint work with James East, Azeef Muhammed and Tim Stokes, we show that the category of DRC-semigroups is isomorphic to a category of certain biordered categories whose object sets are projection algebras in the sense of Jones. This extends the recent groupoid approach to regular $*$-semigroups of East and Muhammed. We also establish the existence of free DRC-semigroups by constructing a left adjoint to the forgetful functor into the category of projection algebras.

Max Gadouleau

Thursday 17th, 10:00

Durham University

Signed graphs and semigroups

Signed graphs are widely used to model networks where interactions are positive or negative. Formally, signed graphs are graphs where the edges are signed positively or negatively. The sign of a cycle is the product of the signs of its edges; a signed graph is balanced if all cycles are positive. In this talk, we'll look at the natural generalisation when the edge signs are taken from a semigroup. We will see that balancing connects with the concept of inverses, and if time permits we'll classify the varieties of balanced signed graphs.

Daniel Heath

Wednesday 16th, 15:00

Free h-adequate semigroups

First considered by Fountain in the 1970s, the class of (right) h-adequate semigroups is a subclass of unary semigroups sitting between ample semigroups and Ehresmann semigroups, and form a natural sister class to birestriction semigroups. These classes are widely studied, having featured in work of many authors including Aird, Branco, Fountain, Gomes, Johnson, Kambites, Kudryavtseva, Lemut Furlani, Szakács and, of course, Gould. It appears however that h-adequate semigroups have fallen through the cracks; here we explore this almost "forgotten" quasivariety. We motivate their study via free adequate semigroups and give a description for the free objects in terms of trees akin to those of Munn for inverse semigroups and Kambites for Ehresmann semigroups.

Michael Kinyon

Monday 14th, 14:45

University of Denver

Quasivarieties of regular $*$-semigroups

A *regular $*$-semigroup* is a regular semigroup with an inversion map $x \mapsto x^*$ which is an involutory antiautomorphism, that is, $(x^*)^* = x$ and $(xy)^* = y^*x^*$. Examples include inverse semigroups, but there are other natural examples as well, such as partition monoids and other diagram monoids.

Idempotents of the form xx^* or y^*y in a regular $*$-semigroup S are called *projections* and the set $P(S)$ of all projections carries a lot of information about the structure of S. Many (quasi)varieties of regular $*$-semigroups can be described in terms of projections. For example, Nordahl and Scheiblich showed that a regular $*$-semigroup is orthodox if and only if it satisfies the projection identity $(pqr)^2 = pqr$ for all $p, q, r \in P(S)$.

In this talk, after a brief general overview, I will describe a couple of interesting quasivarieties of regular $*$-semigroups. The first is the variety of locally inverse $*$-semigroups, which has a nice (and apparently, new) characterization in terms of projections. The second, currently nameless quasivariety is more interesting, and is motivated by an attempt to find a description and representation theorem for the class of regular $*$-semigroups in which partition monoids live.

This is joint work with James East, Matthias Fresacher, Victoria Gould, P. A. Azeef Muhammed, Timothy Stokes, and Abdullahi Umar.

Naftoli Kolodny

Tuesday 15th, 11:30

Binghamton University

The Thompson partition monoid and finitely presented Jónsson-Tarski algebras

The Thompson groups $T \leq V$ were introduced by R. Thompson in the 1960s as the first known finitely presented infinite simple groups. The group V may be regarded, in some sense, as an infinite analogue of the finite symmetric groups. Analogues of V have also been developed for the full transformation monoids, symmetric inverse monoids, and partial transformation monoids; in each case, these structures are finitely presented, infinite, and have no proper quotients. In this talk, we introduce a partition monoid analogue PV of V. One of the many ways of defining V is as the automorphism group of the free Jónsson-Tarski algebra, and in this spirit, elements of PV correspond to isomorphisms between finitely generated subalgebras of finitely presented Jónsson-Tarski algebras. We show that PV is finitely generated but does have proper quotients. We solve the word problem by representing elements as finite machines and give diagrammatic representations of elements analogous to those for partition monoids. We then describe the Green's relations, as well as proving their decidability. In the course of this work, we also develop new results on finitely presented Jónsson-Tarski algebras, including solutions to the word problem, the isomorphism problem, and the subalgebra membership problem.

Valdis Laan

Friday 18th, 10:00

University of Tartu

Enlargements and Morita contexts for semirings

Enlargements were introduced by Mark Lawson in 1996 to study the structure of regular semigroups. Later he proved that two semigroups with local units are Morita equivalent if and only if they have a joint enlargement. For semirings, enlargements can be defined in a very similar way: a semiring S is called an enlargement of its subsemiring R if $S = SRS$ and $R = RSR$. We say that a semiring S is factorisable if $S = SS$. For example, a full matrix semiring over a factorisable semiring R is an enlargement of R.

It turns out that two factorisable semirings are connected by a unitary and surjective Morita context if and only if they have a joint enlargement. Thus, enlargements can be used to investigate Morita equivalence of semirings, similarly to the case of semigroups or rings. In particular, with the help of enlargements one can study Morita invariants — these are properties shared by all semirings in the same Morita equivalence class. It turns out that (von Neumann) regularity is not a Morita invariant for semirings with identity, while it is a Morita invariant for monoids and rings with identity. We will also give some examples of properties that are Morita invariants for semirings.

Ajda Lemut Furlani Wednesday 16th, 14:00

IMFM Ljubljana and University of Ljubljana

F-congruences on F-inverse monoids

A congruence on an F-inverse monoid will be called an F-congruence if it respects the unary operation $^{\mathfrak{m}}$, which maps an element to the maximum element of its σ-class. It is well known that congruences on inverse semigroups are described by kernel normal systems or, equivalently, by congruence pairs. In the talk I will discuss various results concerning F-congruences on F-inverse monoids, including necessary and sufficient conditions for a kernel normal system or a congruence pair associated to an F-inverse monoid S to give rise to an F-congruence on S.

Alex Levine Monday 14th, 14:00

University of East Anglia

Commutative decompositions of (partial) transformation monoids

The commutative submonoid width of a monoid M is the smallest k such that M is a product of k commutative submonoids. We discuss the commutative submonoid width of the symmetric group, the full transformation monoid, the partial transformation monoid and the symmetric inverse monoid, using topological methods to find lower bounds and explicit submonoids to provide upper bounds. Based on joint work with Luna Elliott.

Catherine Reilly Wednesday 16th, 12:00

University of East Anglia

One-relator inverse monoids with decidable word problem are algorithmically unclassifiable

A classical result in combinatorial group theory, the Adian–Rabin Theorem, states that there does not exist an algorithm that takes a finitely presented group G and decides whether G possesses a given Markov property. An example of such a property is having decidable word problem. Consequently, there is no algorithm that takes a finitely presented group and decides whether the group has decidable word problem. On the other hand, Magnus proved in the 1930s that every one-relator group has decidable word problem, so in the one-relator case such an algorithm (trivially) does exist.

In contrast, Gray proved in 2019 that there exists a one-relator inverse monoid with undecidable word problem. A natural question arising from that work is whether it is possible to classify the one-relator inverse monoids with decidable word problem. In particular, one can ask whether there is an algorithm that takes a one-relator inverse monoid and decides whether it has decidable word problem.

In this talk, I will show that no such algorithm exists. That is, the one-relator inverse monoids with decidable word problem are algorithmically unclassifiable.

Aftab Hussain Shah

Thursday 17th, 16:00

Central University of Kashmir, India

Epimorphisms and pseudo varieties of semigroups

For each of the following conditions, we characterize the pseudovarieties of semigroups V that satisfy it: (i) every epimorphism to a member of V is onto; (ii) every epimorphism to a finite semigroup with domain a member of V is onto; (iii) for every epimorphism from a semigroup S into a semigroup T with S in V and T finite, T is also a member of V.

Jamie Smith

Tuesday 15th, 12:00

University of York

Endomorphism monoids of transformation and partition monoids

We describe the structure of the endomorphism monoids of various monoids of transformations and partitions. There are various well known structural similarities and embeddings between these monoids. We show how the similarities between two of these monoids S and T disappear, modify, or lift to $\mathrm{End}(S)$ and $\mathrm{End}(T)$.

Florian Stober

Wednesday 16th, 11:30

University of Stuttgart

The state of membership in semigroups

The membership problem for an algebraic structure asks whether a given element lies in the substructure generated by a given set of elements. For finite semigroups it is PSPACE-complete in the transformation model (Kozen 1977) and NL-complete in the Cayley table model (Jones, Lien, and Laaser 1976). These worst-case bounds hide a lot of structure, though: for restricted classes the problem is often far easier — for semilattices, for instance, it is in AC^0 in both models.

Parametrizing the problem by varieties of semigroups is a standard approach to obtain a more nuanced classification of the complexity. This approach has been successfully applied to aperiodic monoids in the transformation model (Beaudry, McKenzie, and Thérien 1992) as well as inverse semigroups in the Cayley table model and the partial bijection model (Fleischer, Stober, Thumm, and Weiß 2025). However, the landscape of varieties of finite semigroups is much richer than that of aperiodic monoids or inverse semigroups. New varieties appear that have no monoid or inverse semigroup counterpart. This talk surveys which results do transfer, and which gaps between upper and lower bounds remain. Furthermore, we explore the implications of new results on compression in semigroups (Thumm and Weiß 2026) for the membership problem.

Francesco Tesolin

Wednesday 16th, 14:30

Heriot-Watt University

A Schwarz-Milnor lemma for inverse monoids

An étale action of an inverse semigroup generalises a group action on a set. The action forces the set to have structure of a presheaf of sets over the idempotents of the inverse semigroup. The geometric theory of inverse semigroups focuses on the Schützenberger graphs, they are the strongly connected components of the Cayley graph, which are geodesic metric spaces. We observe that the regular action of an inverse semigroup on the Cayley graph is an étale action and that the Cayley graph has the structure of a presheaf of geodesic metric spaces over the idempotents. Using this we prove an analogue of Milnor-Schwarz: showing that a presheaf of geodesic metric spaces on which an inverse monoid acts properly and cocompactly is quasi-isometric to the Cayley graph. We propose étale actions as a tool for geometric inverse semigroup theory.

Jan Philipp Wächter

Wednesday 16th, 10:00

University of Manchester

Inverse semigroups and the isomorphism problem for context-free trees

Context-free graphs were introduced by Muller and Schupp and appear in group theory as the Cayley graphs of context-free (or, equivalently, virtually free) groups. We will motivate in the talk that context-free graphs are tree-like graphs (i.e. quasi-isometric to a tree) with a certain "regularity condition". In fact, this is reflected in a result due to Gray, Silva and Szakács stating that tree-like Schützenberger graphs of inverse semigroups are context-free if the inverse semigroup is finitely presented. More generally, tree-like (and, thus, context-free in the finitely presented case) parts may also appear in more general Schützenberger graphs.

It is the main idea of geometric inverse semigroup theory that algebraic questions often boil down to graph-theoretic questions on the Schützenberger graphs. For example, checking the $\mathcal{D}$- or $\mathcal{J}$-equivalence of two elements is equivalent to checking whether their Schützenberger graphs are isomorphic as non-rooted graphs and whether there are mutual non-rooted homomorphisms between them, respectively. This underlines the need for algorithms for context-free graphs. As such an algorithm, we will discuss how context-free trees can be represented using finite automata and how they can be checked for isomorphism in this representation.

Yanhui Wang

Thursday 17th, 11:30

Shandong University of Science and Technology

Network semigroups

We introduce the class of network semigroups. These are based on networks that extend the notion of a directed graph. This class properly contains the class of graph inverse semigroups. We investigate the structure of network semigroups. We show that two network semigroups are isomorphic if and only if the underlying networks are isomorphic.

Xia Zhang

Tuesday 15th, 16:00

South China Normal University

On reflections of partially ordered semigroups

Embedding a mathematical structure into another structure with better properties is a general and much used idea. Often what we call the "better" structure is the completion of the initial one. Completions of ordered structures have been extensively studied. A nice completion usually defines a reflection arrow in a proper category, so the completion can be seen as a solution of a universal problem. In this talk, reflections of partially ordered semigroups induced by injective hulls, ideals and cuts are presented successively. Meanwhile, the corresponding full reflective subcategories are obtained.

Yayi Zhu

Tuesday 15th, 14:00

University of St Andrews

Presentations for the wreath product $G \wr T_n$ and its ideals

For a group G, the wreath product $G \wr \mathcal{T}_n$ is isomorphic to the endomorphism monoid of a free G-act. The elements are of the form (g_α, α), where g_α is an n-tuple of group elements, and α is a transformation. The transformation component preserves the structure of the $\mathcal{J}$-classes. The chain of $\mathcal{J}$-classes of $\mathcal{T}_n$ is $J_1 < \cdots < J_n$, where J_i consists of transformations of rank i. The $\mathcal{J}$-classes of $G \wr \mathcal{T}_n$ are of the form $\mathcal{J}_i = \{(g_\alpha, \alpha) : \text{rank}(\alpha) = i\}$, and they also form a chain. Furthermore, the ideals of $\mathcal{T}_n$ are of the form $I_m = \{\alpha : \text{rank}(\alpha) \leq m\}$, and the ideals of $G \wr \mathcal{T}_n$ are $G \wr I_m$.

Existing work has established presentations for wreath products of monoids, and the singular wreath product $G \wr (\mathcal{T}_n \setminus \mathcal{S}_n)$, but presentations for a general ideal of $G \wr \mathcal{T}_n$ are not known.

In this talk, I will present results on the relational depth of the ideals $G \wr I_m$ of $G \wr \mathcal{T}_n$, which determines the minimal $\mathcal{J}$-class whose elements must be contained in any presentation for $G \wr I_m$. The proof is based on analysing Cayley table presentations for the ideals of $\mathcal{T}_n$.

Anatolii Zhuchok Thursday 17th, 12:00

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On the structure of trioids

One of the important classes of universal algebras, trioids, was introduced by J.-L. Loday and M. O. Ronco in 2004 in the context of algebraic topology [1]. A trioid is an algebraic system consisting of a set with three binary associative operations satisfying certain axioms. Trioids generalize semigroups and play an important role in trialgebra theory. After recalling the definition of a trioid, we present examples of trioids and discuss their connections with other algebraic structures, including Poisson algebras, Leibniz algebras, dialgebras, dimonoids, digroups, and (strict) n-tuple semigroups. We establish independence of the trioid axioms [2] and construct the free algebra in the variety of trioids [3]. By adapting the decomposition method to trioids, we extend several structural results from semigroup theory to trioids [3]. The main result states that every countable trioid can be embedded into a 2-generated trioid. This generalizes Evans' classical theorem [4], stating that every countable semigroup can be embedded into a 2-generated semigroup.

References.

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